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Effects of the tube diameter on the propagation of helium plasma plume via electric field measurement

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In this work, the dependence of the length of plasma plume, propagation velocity, electric field in the streamer head, and propagation mode transition on the tube diameter varied in the range of 0.07–4 mm is investigated for the first time. The atmospheric-pressure helium plasma plume, ignited by a positive pulsed direct current voltage with a pulse rising time of 60 ns, is confined inside a long glass tube. First, the decreased tube diameter results in the reduction of the length of plasma plume but the growth of aspect ratio of plasma plume. Second, as the tube diameter decreases, the average velocity of the propagation of plasma plume increases first, then reaches a maximum value at tube diameter of 1 mm, and finally decreases for the tube diameter decreasing further. Third, the electric field in the streamer head, determined by the method based on Stark polarization spectroscopy of He 447 nm line, increases monotonically from 9 kV/cm to 20 kV/cm with the tube diameter decreasing from 4 mm to 0.6 mm. Finally, when the tube diameter is further reduced to 0.07 mm, high-speed photography reveals that the propagation mode of the plasma plume transits from the plasma bullet to the continuous plasma column. *Published by AIP Publishing.* [http://dx.doi.org/10.1063/1.4964280]

I. INTRODUCTION

Atmospheric pressure plasma jets (APPJs) have recently attracted considerable interests due to their promising biomedical applications, such as cancer, wounds, skin, and root canal therapies.[1–4] Generally, APPJs are generated inside dielectric tubes first, excited by alternating current or pulsed power supply with driven frequencies of several kHz, and then expanded into ambient air along the noble gas channel. The plasma plume in the open air can be extended to as long as 10 cm or longer.[5] It allows the plasma treatment for 3D complicated geometry. However, short-life-time reactive species including charge particles can be delivered directly to the surface of samples, achieving high treatment efficiency. In addition, APPJs are rich of reactive species (such as ions, O, O_3, OH, NO_X, etc.), and the gas temperature of the plasma plume stays close to or equal to room temperature.[6] These marks of APPJs are very attractive for biomedical applications.

The miniaturization of APPJs, usually realized by the reduction of tube diameter, is desirable for treating smaller targets,[4] such as sterilization of root canal of teeth[3] and single-cellular-level plasma treatment in cancer therapy.[7] To optimize the efficacy of APPJs' treatment, it is essential to investigate the effects of the tube diameter on the properties of APPJs. Recently, several research groups have done great work on it. According to the experimental and simulated results,[8–12] with the decrease in the tube diameter from several mm to tens of μm, the ignition voltage, length of plasma plume, propagation velocity, plasma density, and excitation temperature of helium increase while the gas temperature of the plasma plume remains almost constant. However, because the decreasing tube diameter was accompanied with a fixed gas flow rate in the previous works,[8–12] the spatial distribution of the mole fraction ratio of helium/air along the jet was inevitably varied, which could impose a great impact on the characteristics of APPJs. For example, Ref. 13 found that if the mole fraction ratio of helium/air is smaller than a certain limit, the propagation of the plasma plume would be terminated. Therefore, the influence of the tube diameter on the properties of APPJs remains ambiguous without the consideration of the associated effects of gas components.

To minimise the impact of ambient air, one possible method is confining plasma plumes inside the tubes. Lately, Ref. 14 produced the helium plasma plume within long hollow-core fibers with the aperture in the range of 0.1–2 mm. They observed that the plasma plume with smaller aperture has higher discharge current density and ignition voltage. Nevertheless, the investigation of the influence of tube diameter on the propagation of plasma plume is still lacking. Therefore, the aim of this work is to clarify the effects of tube diameter in the range of 0.07–4 mm on the length of plasma plume, propagation velocity, electric field in the streamer head, and propagation mode transition.

II. EXPERIMENTAL SETUP AND METHODS

Five quartz tubes with a series of inner diameters (ID) of 0.07 mm, 0.6 mm, 1 mm, 2 mm, and 4 mm are used for this study. The wall thicknesses of the five tubes are 0.15 mm for ID = 0.07 mm, 2 mm for ID = 0.6 mm, and 0.5 mm for the others. The schematic of the experimental setup is shown in Fig. 1. A stainless steel needle is severed as the high voltage (HV) electrode and inserted into the tube. The radius of the

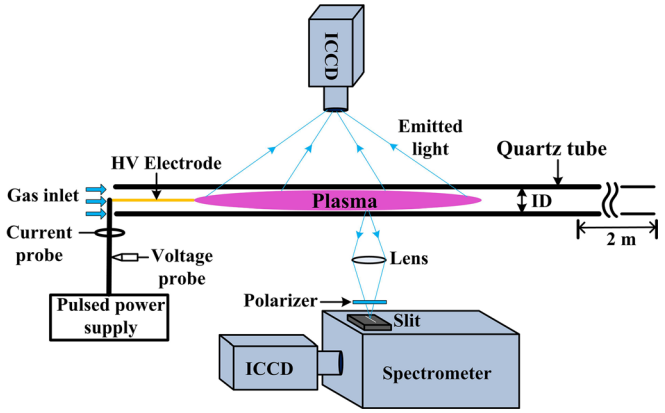


FIG. 1. The schematic of the experimental setup.

needle tip is about 100 μm. Helium gas with purity of 99.999% is fed into the tube, and the gas flow rate controlled by the mass flow meter (MKS M100B) is fixed at 1 l/min throughout this paper. To prevent the ambient air contamination, a 2 m long flexible tube is connected to the right end of the tube. Therefore, the gas component within the discharge area can be reasonably treated as the same with different tube diameters, as reported in our previous work.[15] In addition, it should be pointed out that the electrode configuration for tube ID of 0.07 mm is slightly different. Because the tube ID of 0.07 mm is smaller than the radius of the needle tip, the needle is not inserted into the tube directly but placed inside a relatively large tube with ID of 0.6 mm. Then, the small tube is inserted into the large tube and the joint is sealed for preventing gas leakage. The distance between the needle tip and the end of small tube is set at about 1 mm.

The needle is directly connected to the pulsed dc power supply (Model PVX-4110) with a pulse rising time of 60 ns. When high voltage (typical parameters: amplitude of 6 kV, pulse width of 5 μs, and pulse repetition frequency of 5 kHz) is applied to the needle and helium gas is injected into the tube, an atmospheric-pressure plasma plume is generated within the tube.

A Pearson 2877 current probe and a Tektronix P6015 voltage probe are used to measure the discharge current and the applied voltage, respectively. The diameter of aperture of the current probe is about 6.1 mm, which is much larger than the outer diameter of the high-voltage cable. It allows us to measure the discharge current by letting the cable passing through without physical contact to the probe. The current and voltage waveforms are recorded by a Tektronix DPO3034 broadband digital oscilloscope. A time-resolved emission optical system (OES) based on a PI-MAX2 intensified charge-couple device camera (ICCD) and a PI Acton SpectraHub 2500i spectrometer is used to record the He 447 nm line profile emitted from the plasma plume. The grating and the slit width of the spectrometer are fixed at 3600 grooves/mm and 50 μm, respectively. The instrumental broadening width (full width at half maximum) of the emission spectra line is 0.06 nm. The exposure time of ICCD is fixed at 40 ns, and each fast ICCD image is the integrated picture of 10^6 shots with the same delay time corresponding to the pulse rising edge of applied voltage. ICCD is triggered by the delay pulse from the signal generator (Tektronix AFG3102C) and synchronized with the voltage pulse. A polarizer is located at the front of the slit of the spectrometer, parallel to the propagating direction of plasma plume. Thus, the plasma radiation is polarized in the axial electric field direction (π-polarized).

The electric field in APPJs is measured by the Stark splitting and shifting of He 447 nm line, and its forbidden components. Basic principles have been specifically described in Refs. 16 and 17. In short, three components of the He 447 nm line can be identified by time-resolved OES method: allowed component ($2p^3P^0$–$4d^3D^0$), forbidden component ($2p^3P^0$–$4f^3F^0$), and electric field-free component. Due to the strong dependency of the separation ($\Delta\lambda_{FA}$) between the peaks of allowed component and forbidden component on the electric field strength in the range of 4–30 kV/cm,[17] the electric field can be determined if the value of $\Delta\lambda_{FA}$ is available. In this work, the allowed and forbidden components of He 447 nm line detected by the fast time-gated ICCD mainly originate from the streamer head. Therefore, the electric field strength in the streamer head can be obtained.

III. RESULTS

A. Discharge current and applied voltage waveforms

The I–V waveforms for tube ID of 4 mm and 1 mm are shown in Figs. 2(a) and 2(b), respectively. The discharge current was obtained by subtracting the displacement current from the total current. Two discharge current pulses within a positive voltage pulse were observed, which correspond to the rising and falling edge of the voltage pulse, respectively. The second discharge is generally considered to be related to the residual charges left from the first discharge.[18,19] The peak value of discharge current for the case of ID = 4 mm is 0.083 A, which is 2.6 times of that for the case of ID = 1 mm.

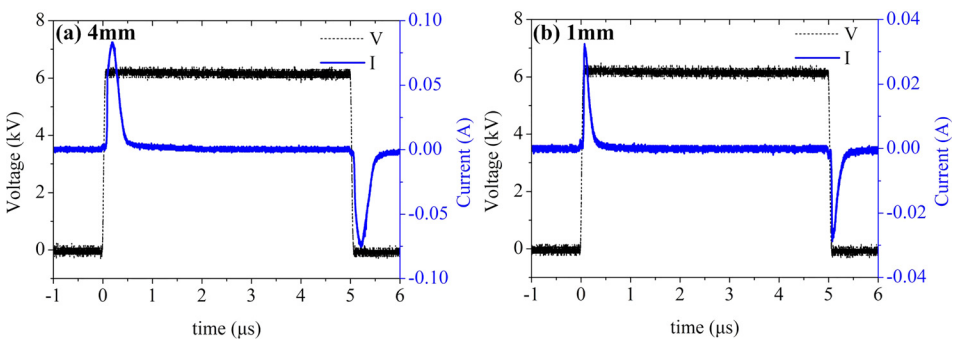


FIG. 2. The discharge current and applied voltage waveforms for tube diameter of (a) 4 mm and (b) 1 mm. Pulse repetition frequency is 5 kHz. The gas flow rate is fixed at 1 l/min.

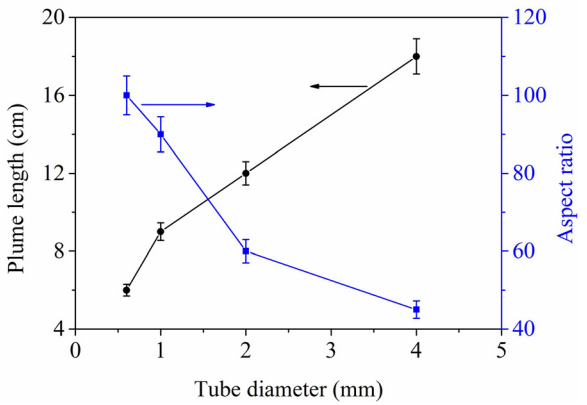


FIG. 3. The length and the aspect ratio (length/diameter) of the plasma plume versus the tube diameter. The parameters of the applied voltage: amplitude of 6 kV, pulse width of 5 μs, and pulse repetition frequency of 5 kHz. The gas flow rate is fixed at 1 l/min.

But the maximum current density of the former is 0.17 times of that of the latter. The pulse widths (full width at half maximum) of the discharge current are 244 ns for ID of 4 mm and 100 ns for ID of 1 mm. It means that the pulse discharge for the former lasts relatively longer.

B. Plume length

The dependence of the length and the aspect ratio of plasma plume on the tube diameter are shown in Fig. 3. As the tube diameter decreases from 4 mm to 0.6 mm, the plume length monotonically decreases from 18 cm to 6 cm. This result is contrary to the simulated results in Refs. 8–10, where the plasma plumes are generated in the open air instead of glass tube. A detailed discussion will be presented in Section IV. In addition, it is interesting to observe that the aspect ratio of the plasma plume increases with the decreasing tube diameter.

C. Propagation of plasma plume

The dynamics of the discharge for different tube diameters were captured by a fast time-gated ICCD. In Fig. 4(a), rather than a continuous plasma column observed by naked eyes, a bullet-like plasma volume is traveling at a high speed of 10^4–10^5 m/s. Similar results have been reported in previous studies.[5,20] After a decade of studies, it is now widely accepted that the "plasma bullets" are ionization wave or streamer guided along the jet.[8,21] Moreover, the propagation velocity of the plasma plume was evaluated by the head positions of the plasma bullet with the knowledge of time difference. The detailed information of propagation velocities for different tube diameters is plotted in Fig. 4(b). It clearly shows that the smaller the tube ID is, the faster the velocity deceases along the axial distance. Interestingly, as the tube diameter decreases from 4 mm to 0.6 mm, the average velocity of the plasma bullet increases linearly first, then reaches a maximum value of 0.7×10^5 m/s at the tube diameter of 1 mm, and finally decreases with the tube diameter further decreasing. This result is unexpected because the longer plasma plume is usually accompanied with higher propagation velocity.[20,21] This inconsistency may be possibly related to the dependence of the tube diameter, instead of applied voltage, gas flow, or electrode configuration.

When the tube diameter is further reduced to 0.07 mm, the dynamic of the discharge is totally different. As shown in Fig. 5, unlike discrete plasma bullets, a continuous plasma column is observed, which is similar to filamentary discharge.[22] In addition, although a high voltage of 8.5 kV is applied, the average velocity of the propagation of the plasma plume is down to 0.2×10^5 m/s, while the length of the plasma plume is as short as 1 cm.

D. Electric field measurements

In order to better understanding how the tube diameter affects the propagation behavior of the plasma plume, the electric field in the streamer head E_h was measured by the method based on the Stark polarization spectroscopy of the He 447 nm line. The emission spectra mainly emitted from the streamer head was detected by the fast time-gated ICCD and synchronized with the voltage pulse. In Fig. 6(a), a typical recorded π-polarized spectrum of the He 447 nm line

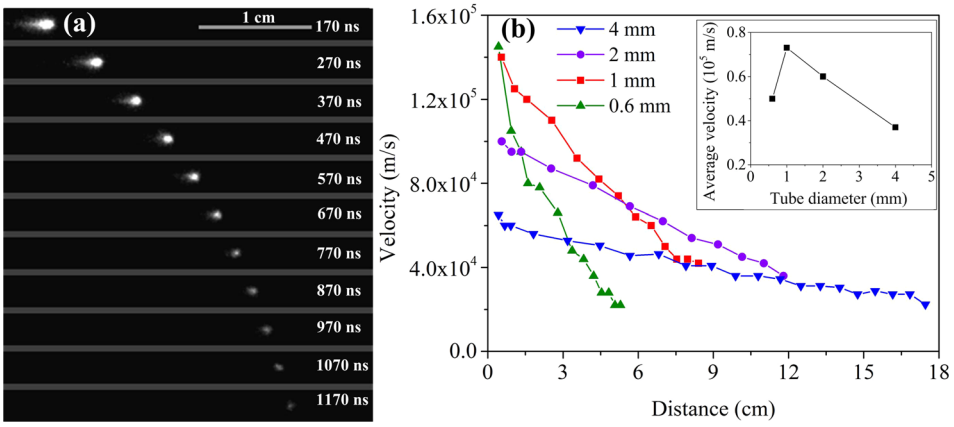


FIG. 4. (a) The high-speed photographs of the propagation of the plasma plume inside the tube with ID = 0.6 mm. The exposure time is fixed at 10 ns. Each image is an integrated picture of 100 shots with the same delay time. The time labeled on each image corresponds to the time in Fig. 2. (b) The propagation velocity versus axial distance for different tube diameters. The zero position represents the needle tip. The inset photograph is the curve of average velocity as a function of the tube diameter. The parameters of the applied voltage: amplitude of 6 kV, pulse width of 5 μs, and pulse repetition frequency of 5 kHz. The gas flow rate is fixed at 1 l/min.

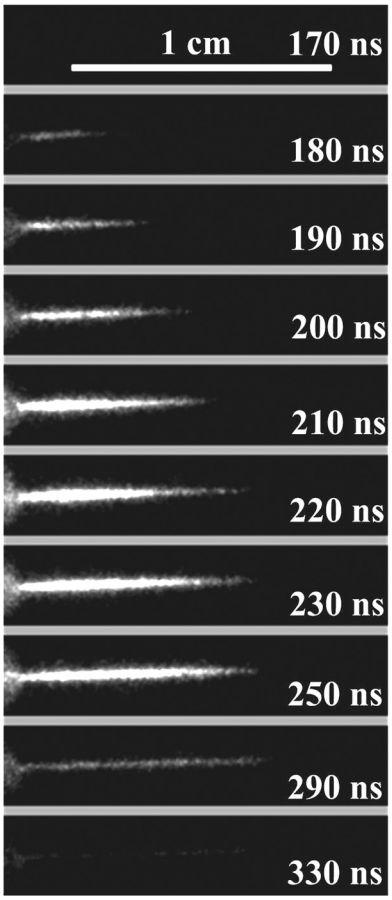


FIG. 5. The high-speed photographs of the propagation of the plasma plume inside the tube with ID of 0.07 mm. The exposure time is fixed at 10 ns. Each image is an integrated picture of 2000 shots with the same delay time. The time labeled on each image corresponds to the time of Fig. 2. Because the decreased tube diameter results in the increased ignition voltage, the amplitude of the pulse voltage is set at 8.5 kV. The pulse width and the repetition frequency are 5 μs and 5 kHz, respectively. The gas flow rate is fixed at 1 l/min.

profile is fitted by Pseudo-Voigt profiles to distinguish the three components (allowed component, forbidden component, and E-free component). The wavelength separation ($\Delta\lambda_{FA}$) between the peaks of allowed line and forbidden line is 0.35 nm, which yields E_h of 19.5 kV/cm. This result is agreed

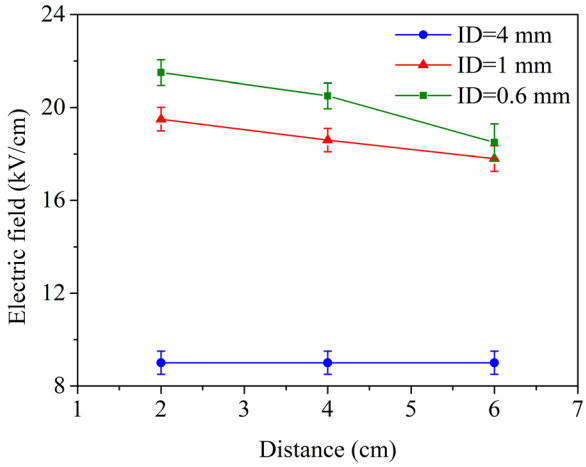


FIG. 7. The electric field strength in the streamer head versus the axial distance for tube ID of 0.6 mm, 1 mm, and 4 mm. The parameters of the applied voltage: amplitude of 6 kV, pulse width of 5 μs, and pulse repetition frequency of 5 kHz. The gas flow rate is fixed at 1 l/min.

well with the simulated and experimental results.[8,10,16,17] Moreover, Fig. 6(b) shows the recorded He 447 nm line profiles for different tube diameters. As the tube diameter decreases from 4 mm to 0.6 mm, both peaks of allowed line and forbidden line shift away from the E-free line centered at 447.14 nm and thus lead to the increase of $\Delta\lambda_{FA}$. In other words, E_h increases as the tube diameter decreases.

The detailed information of the electric field strength as a function of axial distance for different tube diameters was plotted in Fig. 7. First, when the tube diameter decreases from 4 mm to 0.6 mm, E_h increases from 9 kV/cm to over 20 kV/cm. The smaller the tube diameter is, the faster E_h decreases along the axial distance. Furthermore, it should be emphasized that, for the case of tube ID of 4 mm, E_h remains nearly constant during the streamer propagation. All these results are consistent with the propagation velocities in Fig. 4(b), except for the case of tube ID of 0.6 mm. The disagreement lies in that, as the tube diameter decreases from 1 mm to 0.6 mm, E_h increases but the streamer velocity decreases. In addition, it should be mentioned that, for the case of tube ID of 0.07 mm, due to the low signal/noise ratio of the recorded emission spectra, the electric field measurement

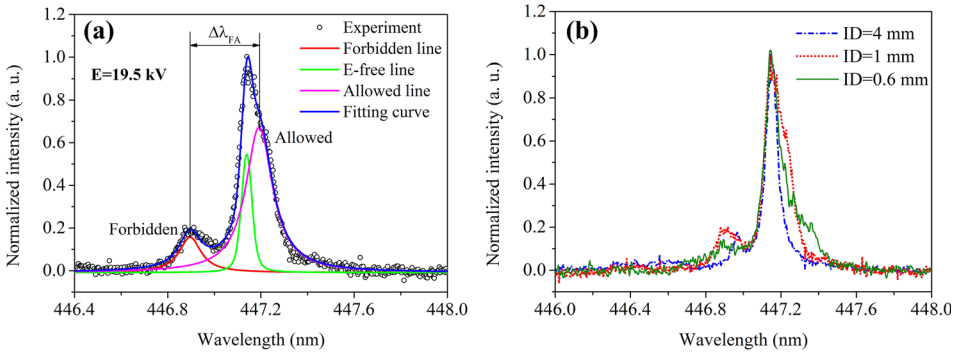


FIG. 6. (a) The Pseudo-Voigt fitting of a typical π-polarized spectra of the He 447 nm line profile for the tube ID = 1 mm. (b) The π-polarized spectra of He 447 nm line emitted from the plasma plume for the tube ID of 0.6 mm, 1 mm, and 4 mm. The grating and the slit width of the optical system are 3600 groove/mm and 50 μm, respectively. The distance between the measured position of plasma plume and the tip of the needle is set at 2 cm. The exposure time of ICCD is fixed at 40 ns, and each image is an integrated picture of 10^6 shots with the same delay time corresponding to the pulse rising edge of applied voltage. The parameters of the applied voltage: amplitude of 6 kV, pulse width of 5 μs, and pulse repetition frequency of 5 kHz. The gas flow rate is fixed at 1 l/min.

based on the Stark polarization spectroscopy of the He 447 nm line is not applicable.

IV. DISCUSSION

A. Effects of the tube diameter on the plume length

In this work, the length of the plasma plume decreases almost linearly with the decreasing tube diameter, which is contrary to the simulation results reported by Naidis[9] and Breden *et al.*[10] This difference is mainly because the plasma plume is generated in the open air, rather than confined inside long tube. Thus, the helium column is shaped by ambient air and determines the plume length.[13] In simulations, for the fixed helium gas flow rate, the decreased tube diameter lengthens the helium column in ambient air, resulting in the increased length of plasma plume. However, in our case, because the plasma plume is generated inside a 2 m long tube, the contamination of ambient air is negligible.

One possible factor is the long-life time species (such as He metastable state), which can provide seed electrons or be ionized relatively easier for the propagation of the plasma plume. Recently, Ji *et al.*[23] observed that the length of the plasma plume generated inside hollow-core-fiber increases with the decreasing diameter of the fiber hole. They believed that the decreased tube diameter results in the increased gas flow velocity, and the long-life time species driven by the gas flow may move further to the downstream area, facilitating the increase of the plume length. However, an opposite result is obtained in our case, and the plume length is weakly dependent on the gas flow rate. Therefore, the long-life time species are expected to play a minor role in the plume length.

Other possible factors are the Reynolds number and pressure difference inside the tube, which are also variable with the tube diameter decreasing. On one hand, Jin *et al.*[24] reported that the length of the He, Ne, or Ar plasma plume inside a glass tube increases with the Reynolds number increasing from 0 to 2000 under laminar flow, while it decreases as the Reynolds number further increases in the laminar-to-turbulent transition region. In this work, the Reynolds number is estimated by using the equation

$$\mathrm{Re} = \frac{\rho_{He} v_g d}{\eta_{He}}, \tag{1}$$

where ρ_{He} is the density of helium gas, η_{He} is the dynamic viscosity of helium gas, and v_g and d are the gas velocity and tube ID, respectively. For the gas flow rate fixed at 1 L/min, the Reynolds numbers are 46, 92, 184, and 306 for the tube ID of 4 mm, 2 mm, 1 mm, and 0.6 mm, respectively. It means that the Reynolds number increases with the decrease of tube diameter, and the gas fluid is always laminar. According to Jin's work, the plume length should increase with the decrease of the tube diameter. However, an opposite result is actually observed in our case. Thus, the effects of the Reynolds number can be neglected. On the other hand, the pressure difference inside the tube is estimated from the Darcy-Weibach equation

$$\Delta P = \frac{\lambda \rho_{He} v^2 L_{tube}}{2d}, \tag{2}$$

where λ is the Darcy friction factor and calculated by $\lambda = 64/\mathrm{Re}$ in laminar, and L_{tube} is the tube length. The calculated pressure differences between the needle tip and the atmospheric pressure with a tube ID of 4 mm, 2 mm, 1 mm, and 0.6 mm are 0 kPa, 0.2 kPa, 2.8 kPa, and 21.5 kPa, respectively. Thus, the pressure difference is too small to be ignored.

Finally, the effect of the tube diameter on the plume length is analyzed from the aspect of energy balance. As we all know, the propagation of the plasma plume is largely dependent on the gained energy (W_g) from the applied electric field and total loss of energy (W_l) due to the ionization, excitation, emitted photons, and heating. The energy balance equation is shown as follows:[25]

$$W_g = W_l + \Delta W, \tag{3}$$

where ΔW is the difference between potential energies with or without plasma plume. In this work, the gained energy from the applied electric field in a single voltage pulse can be roughly estimated by

$$W_g = \int_0^T iV dt, \tag{4}$$

where i is the discharge current, V is the applied voltage, and T is the pulse width of the applied voltage. The gained energy is about 3 μJ for tube ID of 4 mm, which is five times of that for the tube ID of 1 mm. On the other hand, because E_h for the former is much lower than that for the latter, less energy is deposited to ionize the same volume of helium gas. That is possibly why the plasma plume gets shorter with the smaller tube diameter. In addition, it is interesting to observe that the aspect ratio of the plasma plume increases with the decrease of the tube diameter, which just coincides with the results of electric field in Fig. 7.

B. Effects of the tube diameter on the electric field and propagation velocity

Unlike the results reported in Refs. 16, 26, and 27, with the tube diameter decreasing, E_h increases monotonically, while the average velocity of the propagation of the plasma plume decreases after an early peak at tube ID of 1 mm. In 2009, Naidis[26] theoretically revealed that the air streamer velocity increases linearly with the radiation diameter increasing from 0.5 mm to 4 mm. Afterwards, Jansky *et al.*[27] observed a non-linear relationship between the air streamer velocity and the tube diameter, where the streamer velocity reaches a maximum for a tube diameter slightly less than 200 μm with the tube diameter varying in the range of 75–600 μm, which is similar to our case. Lately, Sretenovic *et al.*[16] presented an experimental study on the dependence of the propagation velocity of helium plasma plume on E_h for a single-ring-electrode DBD jet. It denoted that, as E_h increases, the propagation velocity increases first and then

saturates at E_h of 14 kV/cm. These differences are probably related to the different working gas or electrode configurations. In the next, the effects of the tube diameter on E_h and the propagation velocity are analyzed.

According to the analytical model of positive streamer proposed by Kulikovsky,[28] E_h are composed of two components: Laplacian electric field caused by the external applied voltage and local electric field induced by space charge, which is expressed by

$$E_h = E_L + E_\rho = E_L + \frac{e}{\varepsilon_0}\frac{\exp(M)}{4\pi l_\rho^2}, \tag{5}$$

where E_L is the Laplacian electric field, E_ρ is the local electric field induced by the space charge, l_ρ is the width of the space charge layer, M is the Meek's number varying in the range 18–22, ε_0 is the vacuum dielectric constant, e is the electron charge, l_ρ is proportional to the radiation diameter, and the calculated ratio of the radiation diameter to l_ρ varies in the range of 3–5 for the air streamer.[26] In this work, l_ρ is restricted by the tube diameter. Fig. 8 shows that the radiation diameter decreases with the decreasing tube diameter, which means that l_ρ will decrease accordingly and thus leads to the growth of E_h. A similar result was predicted by an isolated head model of the helium streamer.[29] According to the model, when l_ρ decreases from 4 mm to 0.3 mm, E_h increases from 9 kV/cm to 25 kV/cm.

The streamer velocity can be given by

$$v_s = \frac{l_\rho v_h}{\ln(n_h/n_{id})}, \tag{6}$$

where n_h is the electron density in the streamer head, n_{id} is the seed electron density outside the ionization domain, v_h is the ionization frequency in the field E_h, and expressed by $v_h = \mu_e E_h \alpha(E_h)$. Here, μ_e is the electron mobility of 1.13×10^3 cm^2/(V s) at atmospheric pressure in helium, and α is the Townsend ionization coefficient depending on the electric field. First, the streamer velocity is weakly dependent on the ratio n_h/n_{id}. For example, the seed electron density in repetitive helium discharge with the frequency of several kHz is estimated to be 1×10^9 cm^{-3}.[30] Moreover, as the tube diameter decreases from 0.5 mm to 0.08 mm, the electron density of APPJs obtained by Stark broadening of H beta line is in the range of 1×10^{14} cm^{-3} to 8×10^{14} cm^{-3}.[11] Hence, the obtained value of $\ln(n_h/n_{id})$ only varies from 11.5 to 13.6. In other words, the streamer velocity is mainly proportional to the product of l_p and v_h, given by

$$v_s \propto l_\rho E_h \alpha(E_h) \propto dE_h \alpha(E_h). \tag{7}$$

Here, the Townsend ionization coefficients are estimated by

$$\alpha(E) = AN \exp(-BN/E). \tag{8}$$

Here N is the number density of gas molecules in cm^{-3}. A and B are the constants in cm^2 and Td, respectively. The constants AN and BN in pure helium discharge are 920 cm^{-1} and 29.5 kV/cm, respectively.[29] With the knowledge of E_h for a different tube diameter, the values of $dE_h\alpha(E_h)$ can be calculated and listed in Table I. It clearly depicts that the maximum value of $dE_h\alpha(E_h)$ (proportional to streamer velocity) is reached for tube ID of 1 mm, which is consistent with the experimental results in Fig. 4(b). Therefore, the streamer velocity is not only dependent on the electric field but also proportional to the radiation diameter. That is why the streamer velocity decreases even with the increasing E_h as the tube diameter decreases from 1 mm to 0.6 mm.

C. Effects of the tube diameter on the propagation mode transition

When the tube diameter is further reduced to 0.07 mm, high-speed photography shows that the propagation mode of the helium plasma plume transits from plasma bullets to continuous plasma columns, which is similar to filamentary discharge.[22] One possible reason is the increased electric field E_c in the plasma channel behind the streamer head, which could enhance the process of ionization or excitation of atoms or molecules and thus lighten the plasma channel. Also, the increased E_c causes a high voltage drop along the plasma channel, which weakens E_L and thus leads to the fast decrease of E_h with the streamer propagating forward. This is exactly observed in Fig. 7.

Another possible reason responsible for the mode transition is the surface charges accumulated on the inner wall of the tube. Jogi *et al.*[11] pointed out that the characteristic diffusion time of electrons to the walls is reduced to several nanoseconds for a tube diameter less than 0.1 mm. Therefore, for a tube diameter of 0.07 mm, the electrons or ions in the plasma volume are easier to diffuse to the wall of tube, and there charges accumulation or loss through recombination.

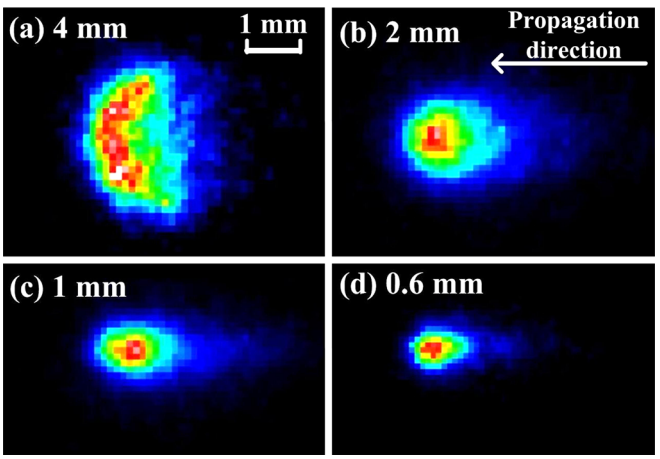


FIG. 8. The high-speed photographs of the helium streamer for different tube diameters. The exposure time is fixed at 10 ns. The parameters of the applied voltage: amplitude of 6 kV, pulse width of 5 μs, and pulse repetition frequency of 5 kHz. The gas flow rate is fixed at 1 l/min.

TABLE I. The values of the tube diameter d, electric field in the streamer head E_h, Townsend ionization coefficients α, and $dE_h\alpha(E_h)$ proportional to the streamer velocity.

d (mm)	E_h (kV/cm)	$\alpha(E_h)$ (cm^{-1})	$dE_h\alpha(E_h)$ (kV/ cm)
4	9	34.7	124.9
1	19.5	202.6	395.2
0.6	21.5	233.3	300.9

These surface charges will cause distortion of the Laplacian electric field or provide seed electrons for the propagation of the plasma plume, which has been proved to be important in dielectric barrier discharges.[31,32] Therefore, the surface charge accumulated on the inner walls might play an important role in the propagation mode transition of the plasma plume. However, there is a lack of enough consistent information to evaluate the effects of surface charges on the propagation of plasma plume, which will be studied in the near future with tools of surface charges measurements for micro-tubes.

V. CONCLUSION

The tube diameter plays a significant role in the propagation behavior of the helium plasma plume confined inside tubes. First, the smaller the tube diameter is, the shorter the plasma plume is, while the higher the aspect ratio of the plasma plume is. One possible reason is that the decreased tube diameter aggravates the imbalance of the gained energy from the external electric field and the loss of energy deposited in the ionization, excitation, and gas heating. Second, as the tube diameter decreases, E_h increases monotonically, while the propagation velocity of the plasma plume decreases after an early peak at tube diameter of 1 mm. This is partly because the decreased tube diameter results in the decreased width of the space charge layer. Third, when the tube diameter is further reduced to 0.07 mm, the propagation mode of the plasma plume converts from the plasma bullet into the continuous plasma column. One possible factor for the mode transition is the increased electric field in the plasma channel, while the surface charge accumulation on the inner wall of tube might also affect the propagation behavior of the plasma plume. Finally, this study gives new insight into the influence of tube diameter on the propagation of helium plasma plume and offers options for guiding the dynamics of the plasma plume with respect to various applications, such as inner surface modification of micro-tubes or developing plasma jets for biomedical applications.

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